

Data Semantics – 25–26 – Exercises on RDF +/- LLMs

1. Extraction of triples from natural language text – manual approach

Goal: learn how to interpret the information in a text through the lens of the RDF graph model.

Activity: analyze the following texts and extract a set of RDF triples in Turtle syntax that adequately represent their content.

1.1. Brown Sugar – Wikipedia abstract ([link](#))

***Brown Sugar** is the debut studio album by the American musician [D'Angelo](#), released on July 3, 1995, through [EMI](#).*

1.2. Voodoo - Wikipedia abstract - ([link](#))

***Voodoo** is the second studio album by American musician [D'Angelo](#), released on January 25, 2000, through [Virgin Records](#).*

2. Extraction of triples from natural language text with an LLM

Context: you are experimenting with an approach that enables the creation of a knowledge graph about musicians and their works, using RDF in Turtle format. The goal of the application you are developing is to build a coherent graph that makes appropriate use of widely shared vocabularies and data formats, in order to facilitate open access to information on the Web.

Goal: learn to critically analyze the results in light of what we have covered in the course.

Activity: use an LLM-based assistant—e.g., ChatGPT—to automatically extract triples from text; use your own judgment to evaluate the quality of the results and identify shortcomings.

Approach: basic method loosely inspired by [an article on RDF extraction via LLMs](#), with two processing steps plus some analysis prompts.

// Texts to use in the prompts

2.1. Brown Sugar – Wikipedia abstract ([link](#))

***Brown Sugar** is the debut studio album by the American musician [D'Angelo](#), released on July 3, 1995, through [EMI](#).*

2.2. Voodoo - Wikipedia abstract - ([link](#))

***Voodoo** is the second studio album by American musician [D'Angelo](#), released on January 25, 2000, through [Virgin Records](#).*

2.3.D’Angelo – Italian Wikipedia abstract ([link](#)), translated and slightly modified (note: the text we submit contains an error that we will return to later)”

D’Angelo, born Michael Eugene Archer (Richmond, February 11, 1974 – New York, October 14, 2025), was an American singer, multi-instrumentalist, songwriter, and record producer.

He is considered one of the founders of modern neo-soul music, a movement that emerged in the mid-1990s and organically blends the sound of hip hop with that of R&B. Throughout his career, he received four Grammy Awards and collaborated with artists such as Lauryn Hill, Erykah Badu, Raphael Saadiq, Common, Dr. Dre, and Jay-Z.

He released two studio albums, *Brown Sugar* (1995) and *Voodoo* (2000), the latter ranked 28th on *Rolling Stone*’s list of the 500 greatest albums. In 2014 he released his third and final album, *Black Messiah*. He died on October 15, 2025.

// Prompt Templates

2.A - Extraction of triples in JSON format / note: the text we submit contains an error.

Extract triples from the text you are given. Return the triples in JSON format.

Here are some examples:

Text: *The Amazon River flows through Brazil and Peru.*

Your answer: {"Triples": [{"Amazon River", "country", "Brazil"}, {"Amazon River", "country", "Peru"}]}

Text: *The symptoms of COVID-19 include fever, cough, and respiratory difficulties.*

Your answer: {"Triples": [{"COVID-19", "symptom", "fever"}, {"COVID-19", "symptom", "cough"}, {"COVID-19", "symptom", "respiratory difficulties"}]}

Text: *The American Civil War took place from 1861 to 1865.*

Your answer: {"Triples": [{"American Civil War", "start date", "1861"}, {"American Civil War", "end date", "1865"}]}

Text:

[[REPLACE HERE]]

Your answer:

2.B - Transformation from JSON to RDF in Turtle syntax

Transform the triples into RDF using the Turtle format.

3. Validation and assessment of the triples automatically extracted by the LLM

3.A - Verify inconsistencies

Check whether there are inconsistencies or conflicting triples among the extracted texts.

3.B - Validate and analyze the triples extracted with the LLM

Questions to consider:

- Are all the triples well-formed and compliant with Turtle syntax?
- Are shared vocabularies being used—where yes, and where not?
- Are the IRIs designed according to a coherent logic, allowing information extracted from different texts to be linked?
- Are the literals organized, or are they all just “plain literals”?
- What happens if we need to merge triples extracted from different texts?
- What happens if I ask to structure the information into one or more tables for a relational database?
- How does Schema.org suggest modeling this information compared to the LLM’s proposals? Has it made correct use of the predicates (classes and properties)?

3.C - Automatic assessment by an LLM

Are there any other issues related to the quality of the representation in the turtle representation above?

If yes, clearly present these issues as a list. Also, propose a solution for each issue

4. Play around with integration of graph data and more challenging modeling

Goal: to verify how the extraction across different texts makes it possible to build a cohesive graph. Here, some freedom is left to individual creativity. You can still try or begin with a fairly simple prompt such as the one below:

4.A - Union of triples extracted from two texts (e.g., D'Angelo and Voodoo)

Now let’s merge these triples related to the album *Voodoo* with those extracted from the abstract about D’Angelo. Identify any redundant triples and decide which ones to keep.

5. Extension of the triples extracted with an LLM and more complex modeling patterns – Extra

- Write two pairs (*text*, *triples*) in which **blank nodes** are used.
- What happens if we want to associate the publication date with the **relationship between the work and the artist**? (see the exercises on **reification** in the support material)